

Nearest Neighbour Algorithms

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Supervised learning

- ▶ Given labelled training examples $(x_1, y_1), \dots, (x_T, y_T)$ where
- ▶ $x_t \in X$ are **features**
- ▶ $y_t \in Y$ are **labels**..

Feature space $\mathcal{X}$

- ▶ Usually $\mathcal{X} = \mathbb{R}^n$: the n-dimensional Euclidean space
- ▶ How do we use your class data?

Classification

- ▶ $Y = \{1, \dots, m\}$ are **discrete** labels

Regression

- ▶ $Y = \mathbb{R}^m$ are **continuous** values

The kNN algorithm idea

- ▶ Assume an unknown example is similar to its neighbours
- ▶ Smoothness allows us to make predictions

Discriminatory analysis-nonparametric discrimination: consistency properties, Evelyn Fix and Joseph L. Hodges Jr, 1951.

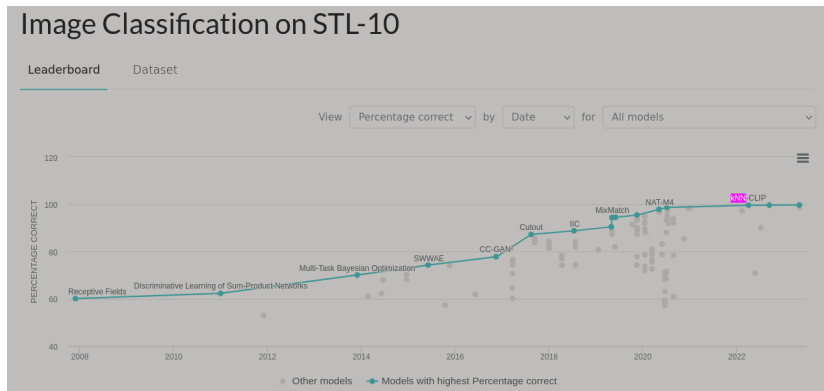


Figure: Evelyn Fix



Figure: Joseph Hodges

Performance of KNN on image classification



- ▶ Really simple!
- ▶ Can outperform really complex models!

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The Nearest Neighbour algorithm

Pseudocode

- ▶ Input: Data $(x_t, y_t)_{t=1}^T$, test point x , distance d
- ▶ $t^* = \arg \min_t d(x_t, x)$ // How do we implement this?
- ▶ Return $\hat{y}_t = y_{t^*}$

Classification

$$\hat{y}_t \in [m] \equiv \{1, \dots, m\}$$

Regression

$$\hat{y}_t \in \mathbb{R}^m$$

Nearest Neighbour Algorithms

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The Nearest Neighbour algorithm

Pseudocode

- Input: Data $\{x_i, y_i\}_{i=1}^n$, test point x , distance d
- $i^* = \arg \min_i d(x_i, x)$ // How do we implement this?
- Return $\hat{y}_i = y_{i^*}$

Classification

$\hat{y}_i \in [m] = \{1, \dots, m\}$

Regression

$\hat{y}_i \in \mathbb{R}^m$

Explanation of $\arg \max$. If $S = \{x_1, \dots, x_n\}$ is a set of elements with the $>$ and $=$ relations, then:

- $\max S \geq x_i \quad \forall x_i \in S$.
- $\exists x_i \in S$ s.t. $\max S = x_i$.

In addition

- $\arg \max S$ are the **indices** i for which $\max S = x_i$.

The k -Nearest Neighbour algorithm

Pseudocode

- Input: Data $(x_t, y_t)_{t=1}^T$, test point x , distance d , neighbours k

The k -Nearest Neighbour algorithm

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- ▶ Calculate $h_t = d(x_t, x)$ for all t .

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Classification

- ▶ We use a **one-hot encoding** $(0, \dots, 0, 1, 0, \dots, 0)$, with $y_t \in \{0, 1\}^m$.
- ▶ The class of the t -th example is $j \Leftrightarrow y_{t,j} = 1$.
- ▶ Equivalently, return p with

$$p_i = \sum_{t=1}^k \mathbb{I}\{y_{s_t} = i\} / k$$

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Regression

- ▶ $y_t \in \mathbb{R}^m$, so we need do nothing

The number of neighbours

Discuss in groups of 2-3 people. Write down your answers.

$$k = 1$$

- ▶ How does it perform on the training data?
- ▶ How might it perform on unseen data?

$$k = T$$

- ▶ How does it perform on the training data?
- ▶ How might it perform on unseen data?

Distance function

For data in $\mathbb{R}^n$, p -norm

$$d(x, y) = \|x - y\|_p$$

Scaled norms

When features having varying scales:

$$d(x, y) = \|Sx - Sy\|_p$$

Or pre-scale the data

Complex data

- ▶ Manifold distances
- ▶ Graph distance

Distances

A distance $d(\cdot, \cdot)$:

- ▶ Identity $d(x, x) = 0$.
- ▶ Positivity $d(x, y) > 0$ if $x \neq y$.
- ▶ Symmetry $d(y, x) = d(x, y)$.
- ▶ Triangle inequality $d(x, y) \leq d(x, z) + d(z, y)$.

For data in $\mathbb{R}^n$, ℓ_p -norm

$$d(x, y) = \|x - y\|_p$$

Norms;

A norm $\| \cdot \|$

- ▶ Zero element $\|0\| = 0$.
- ▶ Homogeneity $\|cx\| = c\|x\|$ for any scalar a .
- ▶ Triangle inequality $\|x + y\| \leq \|x\| + \|y\|$.

p -norm

$$\|z\|_p = \left(\sum_i z_i^p \right)^{1/p}$$

Neighbourhood calculation

If we have T datapoints

Sort and top K .

- Requires $O(T \ln T)$ time

Use the Cover-Tree or KD-Tree algorithm

- Requires $O(cK \ln T)$ time.
- c depends on the data distribution.

Making a decision

kNN as a model

- ▶ Given features x , we get a vector p of class probabilities:

$$p_i = P(y = i|x),$$

where $P(y = i|x)$ is the probability that y is i , given x .

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Decisions to maximise accuracy

At time t :

- ▶ We observe features x_t

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The model versus the prediction

- ▶ The **model** P tells us the probability of different classes.
- ▶ When we **decide** what our prediction should be, we can **use** the model.
- ▶ We will use π to denote the **decision rule** or **policy**.

Decisions versus predictions

- ▶ We frequently need to make a **decision**, instead of just a **prediction**.
- ▶ Our **utility** function $U(y, a)$ represents our **preferences**.
- ▶ The space of **actions** A is *not identical* to the set of **labels** Y .

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Minimise spam annoyance

What utility function would you use for the spam detection problem?

Utility	Pass	Flag	Trash
Normal			
Spam			
Virus			

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Classification decision to maximise expected utility

- ▶ Expected utility of a single decision

$$\mathbb{E}[U|a, x] = \sum_y P(y|x, a) U(y, a) = \sum_y P(y|x) U(y, a)$$

- ▶ The decision maximising expected utility: $a^* = \arg \max_a \mathbb{E}[U|a, x]$

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Setup

- ▶ Use the class dataset from earlier (or another dataset given in class).
- ▶ Use scikit-learn's `KNeighborsClassifier` (or `KNeighborsRegressor` if your target is continuous).

Tasks

- ▶ Fit the model for several values of k (e.g. $k = 1, 3, 5, 15, T$), recording both **training** and **test** accuracy for each.
- ▶ Rescale the features (e.g. with `StandardScaler`) and repeat. Does the ranking of neighbours change? Does performance change?
- ▶ Instead of rescaling the features, change the **distance** function. Can you change it so it has the same effect?

Setup

As a class, agree on a utility matrix $U(y, a)$ for the spam example from the slides, by filling in the table:

Utility	Pass	Flag	Trash
Normal			
Spam			
Virus			

Tasks

- ▶ Fit a kNN classifier and call `predict_proba()` to get $P(y | x)$ for each test point.
- ▶ Write a function `utilityScore(y, actions, U)` that takes the true labels y_t and chosen actions a_t , and returns $\sum_{t=1}^T U(y_t, a_t) / T$.
- ▶ Write a second function `utilityAction(probs, U)` that returns, for each test point, the action a^* maximising expected utility under $P(y | x)$ and your U .
- ▶ Compare the average utility of plain `predict()` (most likely class) against your `utilityAction()`. Which one wins? Does the answer depend on which U you chose?
- ▶ Once you have $a^* = \arg \max_a \sum_y P(y|x) U(y, a)$, **prove** that no other decision rule can achieve higher expected utility under your model.
- ▶ Show that if $A = Y$ and if $U(y, a) = \mathbb{I}\{y = a\}$ (you only care about being right), then a^* is the most likely class, $\arg \max_y P(y | x)$.
- ▶ Bonus: What changes once $P(y|x)$ is **estimated** (e.g. via kNN) rather than known exactly? For example, say that $P^*(y|x)$ is the true model.

Behaviour of k (pen and paper)

- ▶ Argue why the **training** accuracy of k -NN is always exactly 100% when $k = 1$.
- ▶ Argue why training accuracy can only get **worse, or stay the same**, as k grows from 1 towards T . (Hint: what does the prediction become when $k = T$?)
- ▶ Does the same argument apply to **test** accuracy? Why, or why not? Check this against the plot you made in Activity 1.

Average behaviour of k

- ▶ Calculate the accuracy for 100 different synthetically generated training and testing datasets
- ▶ Average the training and testing accuracy over these 100 runs.
- ▶ What do you see?

Deciding from a generic utility

Let $U(y, a)$ be an arbitrary utility matrix and $P(y | x)$ an arbitrary model.

- Write down the expected utility of action some a .